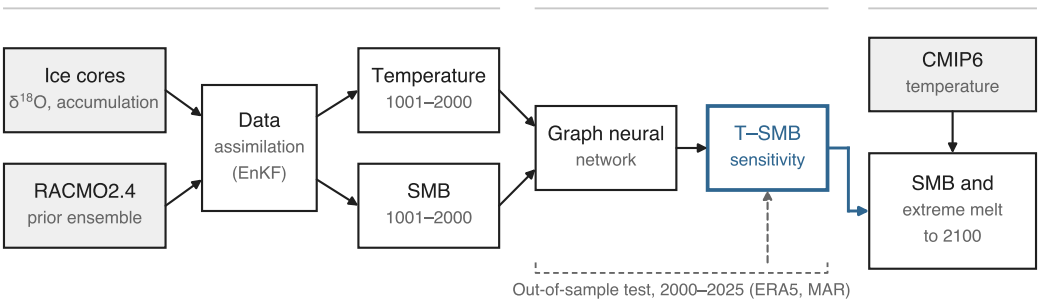
1 Reconstruct
2 Learn
3 Project
Ice cores
δ18O, accumulation
RACMO2.4
prior ensemble
Data
assimilation
(EnKF)
Temperature
1001–2000
SMB
1001–2000
Graph neural
network
T–SMB
sensitivity
CMIP6
temperature
SMB and
extreme melt
to 2100
Out-of-sample test, 2000–2025 (ERA5, MAR)